

How not to explode (maybe)

Manual for the game *Defuse or Lose*
created at PoliJam 2025

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Introduction to bomb defusal

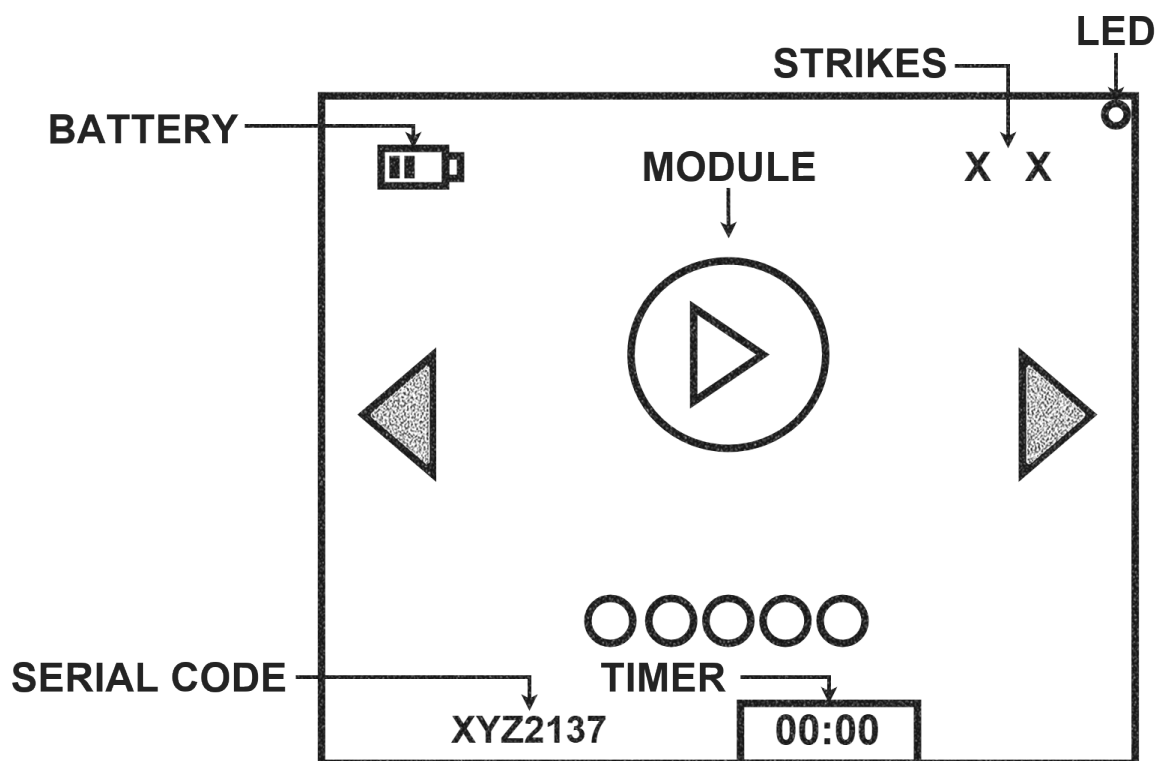
This guide describes how to defuse a bomb once the defusing device has been connected to it (that process is covered in our other manual, "*How to connect the defusing tool to a bomb and not explode (maybe)*").

The bomb explodes when the timer shown on the device reaches 0 or when the defuser makes three mistakes.

The device has modules, and each module contains a puzzle. Solving the puzzles in the modules defuses the bomb. You move between modules with the arrows shown on the device. The number of modules and their status (whether they have been completed) is shown at the bottom of the device screen.

Remember! Switching to another module resets the puzzle (for some modules it may even change its rules).

Note! Modules may make use of the elements of the bomb. Here they are:



Hazards

If defusing a bomb is not dangerous enough for you, we have good news! The defusing device may change its behaviour depending on the surrounding conditions.

Train:

Riding a train causes shaking, which does not exactly help while defusing a bomb. On top of that, every now and then the train enters a tunnel, and the carriage you are in turns as black as night (good luck reading the serial code!).

Storm:

During a storm, flashes of lightning brighten the darkness that falls when the train enters a tunnel. However, the defusing device may act up... Electrical discharges temporarily scramble the elements of the bomb, making it impossible to read elements such as the battery, the timer or the number of strikes.

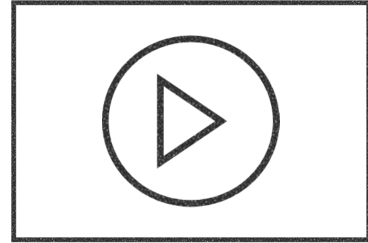
Fallout:

Every now and then the train passes through a radioactive area. The radiation gives the defuser some unpleasant side effects... Being unable to control the defusing device for a moment is probably the least of their worries.

Hazards appear gradually, as the defuser gains experience. Beginner defusers only work during a train ride, while professionals are not bothered by all the hazards being present at once.

Button

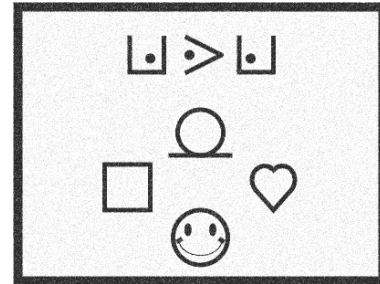
The Button module is about pressing a button (I bet you have already figured that out). Each button has to be pressed differently, and how to press a given button depends on the icon it displays.



1. If the button shows a Play icon (a triangle), press and hold the button for as many seconds as there are bars on the battery.
2. If you see a Stop icon (two rectangles side by side), press the button as many times as the sum of the digits in the bomb's serial code. If the serial code contains no digits, press the button once.
3. For the Record icon (a capital letter R), press the button and release it only when the timer contains the digit 4.
4. When the button shows a heart, do nothing for 15 seconds (you may breathe).

Signs

The Signs module displays four symbols, and the patterns above them hint at which one to press. Each instruction is an encrypted code - it may refer to the name of a symbol or to the position of the button to click. To defuse the bomb, this process has to be repeated four times.



Above the symbols there is a cipher. Once it is decoded, we learn which symbol, or the button at which position, has to be clicked. To decrypt the code, use the tables below. Each sign is a shape cut out of a table, and that piece holds a letter. The dot tells you which of the letters should be used.

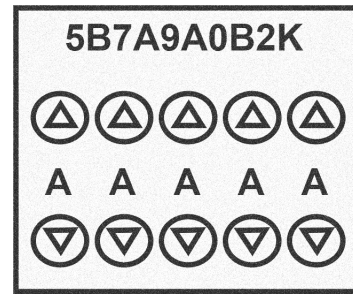
I am hungry while writing this description, so I will explain it to you using boxes and cookies. Each sign is a box with a cookie locked inside, and the cookie shows from which side we should reach for it, which leads us to the position of the letter in one of the tables. Some cookies come from boxes in the table on the left, and some from the table on the right.



T N	I H	S C
A O	P M	E F
R Z	D G	W L

Cipher

The Cipher module is about decoding secret government information. The digit in front of a letter tells you how to decode it, and the letter itself changes according to a specific rule (you, on the other hand, never change).

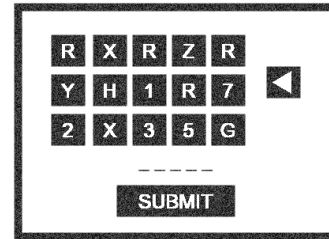


We split the cipher into sections of 2 characters: a digit and a letter. Decrypting is a piece of cake, just go step by step: start with the digit and check which range it belongs to, then check whether the digit is even or odd, and finally find the letter from the cipher and swap it for the matching one in the table below. Well done, you have just decrypted the first letter!

<u>0-4</u>		<u>5-9</u>	
<u>Even</u>	<u>Odd</u>	<u>Even</u>	<u>Odd</u>
A → X	C → J	A → N	C → Q
B → Y	J → K	B → O	J → B
D → W	K → L	D → U	K → C
G → T	L → Z	G → V	L → D
F → M	Z → R	F → S	Z → A
H → P		H → E	

Keyboard

In the Keyboard module the defuser has to share information about their device with the bomb. The Keyboard module works differently depending on the number of strikes. The parts of the answer are typed without spaces.



If there are no strikes, the code to enter is:

- User name - the first 6 characters of the defuser's user name in the operating system. If the name is shorter, write it in full. If the defuser does not know their own name, they can type "echo %USERNAME%" in cmd.
- Defusal date - the day and month in which the bomb is being defused (remember that the date must match the defuser's time zone... unless you are cheating and reading this manual while defusing the bomb). The date is written in DDMM format.

With one strike, the code looks like this:

- Computer name - the first and the last character of the device name in the system (names can be long, luckily the bomb only checks these two characters). You can find it in "This PC" or with "echo %USERDOMAIN%".
- Operating system - if the defuser uses Windows, they should type "windows"; for any other system type "linux" (bomb makers do not know much about operating systems).

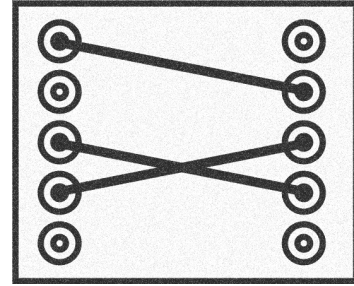
If the defuser has two strikes, then first of all you two should work on your communication, and second of all the code consists of the answers to three questions. The possible answers are "yes" or "no".

- System uptime - the defuser types whether their device has been running for more than an hour.
- RAM - if the defuser's computer has at least 8 GB of RAM, it has enough RAM to run Google Chrome. Only then may they type "yes".
- Graphics card - if the defuser uses an NVIDIA graphics card, they should type "yes".

*If the module rejects a correct answer, it may have read the device data incorrectly. If the module does not work properly because of that, it can be skipped by entering "xxxxxxxxxx" as the answer.

Cables

This module simulates connecting the wires found in the bomb (these wires are not real). To solve the puzzle you have to connect 5 pairs of wires, numbered from top to bottom. Not all wires have to be connected - unless you want to explode.



If the battery is at about 25%, connect the first left slot with a wire to the right slot that is fourth from the end.

If the battery is fully charged, connect the second left slot to the last right slot.

If the last character of the serial code is a digit, and an even digit at that, connect the third left slot (or let the defuser connect it) to the middle right slot.

Calculate the sum of the character values in the serial code - digits as they are, letters as their positions in the alphabet (see the table at the bottom of the page). Divide the sum by 5 - the remainder gives the right slot for the second-to-last left slot (if the remainder is 0, treat it as 5).

If the serial code contains 3 or more vowels, connect the first left slot to the fourth right slot, as long as the first left slot has no connection assigned yet.

If the first character of the serial code is a consonant and the second left slot has no connection assigned yet, connect it to the third right slot.

If, after checking all the rules above, any left slot is still unassigned (except the fifth one), connect it to the first free right slot in order (repeat if necessary).

To calculate the sum of the serial code faster, the letter values are listed below.

A → 1	G → 7	N → 14	U → 21
B → 2	H → 8	O → 15	V → 22
C → 3	I → 9	P → 16	W → 23
D → 4	J → 10	Q → 17	X → 24
E → 5	K → 11	R → 18	Y → 25
F → 6	L → 12	S → 19	Z → 26
	M → 13	T → 20	

Signal

This module is responsible for receiving a signal from headquarters. The bomb receives the signal and plays it back to the defuser with a red LED, using Morse code.



The LED plays the signal in a loop - first the first digit, then the second one (unless the signal is a single-digit number). Between the digits of the number the signal pauses for two seconds, and after the whole number has been played the LED pauses for four seconds.

Remember not to switch modules just to play the signal from the beginning! Headquarters sends a different signal every time the module is started.

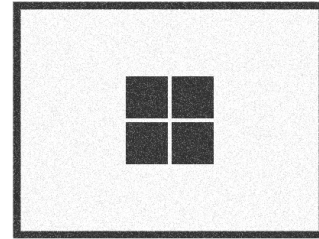
The Morse code values are shown in the table below. The "-" sign (dash) stands for a long signal, the "•" sign (dot) stands for a short signal.

Digit	Morse code
0	- - - - -
1	• - - - -
2	• • - - -
3	• • • - -
4	• • • • -

Digit	Morse code
5	• • • • •
6	- • • • •
7	- - • • •
8	- - - • •
9	- - - - •

Memory

The Memory module tests your ability to memorise sequences and react quickly. While the module is running, successive stages appear, and each of them requires not only following the instruction precisely, but also recalling the results of the previous rounds.



At the start of each stage one of the buttons goes dark - depending on which one went dark, you have to press the right button as the answer.

You are probably wondering what the abbreviations TL, TR, BL and BR mean, so let me help. They are button positions: TL (Top-Left) is the top left button, TR (Top-Right) is the top right one, BL (Bottom-Left) is the bottom left one, and BR (Bottom-Right) is the bottom right one. In case you are curious what "the button opposite" means, it means exactly what you think: if you have to press the button opposite TL, the correct choice is BR (I did not think I would have to explain that).

Stage 1:

If the TL button flashes, press the TR button.

If the TR button flashes, press the BL button.

If the BL button flashes, press the BR button.

If the BR button flashes, press the TL button.

Stage 2:

If the TL button flashes, press the button that flashed in stage 1.

If the TR button flashes, press the button that was pressed in stage 1.

If the BL button flashes, press the button opposite the one that flashed in stage 1.

If the BR button flashes, press the same one that flashed in stage 1.

Stage 3:

If the TL button flashes and the player has at least 1 strike, press the one that flashed in stage 2; otherwise press the one that flashed in stage 1.

If the TR button flashes, press the one that was pressed in stage 1.

If the BL button flashes, press the one that flashed in stage 2.

If the BR button flashes, press the button opposite the one pressed in stage 1.

Stage 4:

If the TL button flashes, press the one that was pressed in stage 1.

If the TR button flashes, press the one that was pressed in stage 2.

If the BL button flashes, press the one that flashed in stage 3.

If the BR button flashes, press the button opposite the one pressed in stage 2.

Stage 5:

If the TL button flashes, press the one that flashed in stage 1.

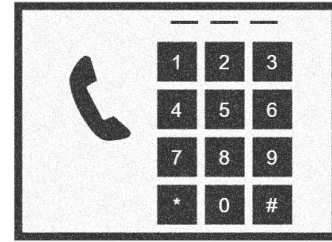
If the TR button flashes, press the one that flashed in stage 2.

If the BL button flashes, press the one that flashed in stage 4.

If the BR button flashes, press the one that flashed in stage 3.

Entryphone

The Entryphone module is about picking the correct flat number based on the sounds coming from it. You can hear the sounds from the flat by using the button with the handset (the sounds will be low quality - the defusing device is not an iPhone).



The flat number can be calculated from the order in which the sounds are played. The sound in the first position sets the starting value, while the sounds in the second and third positions tell you which calculations to perform on the previous value.

The answer to the puzzle must be a positive whole number - if the result has digits after the decimal point, drop them, and if it is a negative number, take its absolute value.

The "*" key on the entryphone keypad confirms the answer, while "#" resets it.

Opening a can:

Position 1:
Start with 8.

Position 2:
Add 4.

Position 3:
Multiply by 2.

Bluetooth connection:

Position 1:
Start with 12.

Position 2:
Subtract 6.

Position 3:
Add 5.

Car starting:

Position 1:
Start with 15.

Position 2:
Multiply by 2.

Position 3:
Divide by 5.

Cat meowing:

Position 1:
Start with 20.

Position 2:
Subtract 4.

Position 3:
Multiply by 3.

Eating crisps:

Position 1:
Start with 6.

Position 2:
Add 3.

Position 3:
Divide by 2.

Doorbell:

Position 1:
Start with 10.

Position 2:
Multiply by 5.

Position 3:
Subtract 7.

Windows starting up:

Position 1:
Start with 30.

Position 2:
Divide by 3.

Position 3:
Add 15.